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SOFTWARE

Instrumented systems

Crawlers, pipelines, search indexes, reporting layers. Built around the actual data model: queryable; auditable; extendable.

Automation · Search · Data Pipelines · CLI · Reporting

Workflow automation

Manual processes converted to defined pipelines: structured inputs, quality gates, clean handoffs.

Automation · Search · Data Pipelines · CLI · Reporting

SYSTEMS

Architecture before components

Topology, constraints, and failure modes defined first.

Topology · Process Design · Constraints · Documentation · Operations

Process formalization

Informal rules surfaced, named, and documented.

Topology · Process Design · Constraints · Documentation · Operations

DESIGN

DC lighting architecture

Distribution layout, driver selection, control protocol, dimming behavior, serviceability.

Low Voltage · DC Distribution · Typst · Specification · Diagrams

Technical publication

Specifications, diagrams, and installation logic precise enough to build from. (Clear enough to hand to a non-specialist without a walkthrough.)

Low Voltage · DC Distribution · Typst · Specification · Diagrams

RESEARCH

Primary source analysis

Datasheets, standards, supplier catalogs, field constraints.

Standards · Supply Chain · Specs · Field Constraints · Technical Writing

Actionable synthesis

Gap between specification and real installation conditions: identified, closed, documented.

Standards · Supply Chain · Specs · Field Constraints · Technical Writing

IMPLEMENTATION

Full-cycle execution

Layout, sourcing, commissioning, verification – without coordination loss from splitting phases across specialists who don't share a system model.

Installation · Commissioning · Prototyping · Verification · Low Voltage

Constraint-driven prototyping

Early builds targeting the hard problem: thermal behavior, dimming continuity, distribution loss, control latency. Constraint solved before the system is committed.

Installation · Commissioning · Prototyping · Verification · Low Voltage

ENGAGEMENT

Where this applies

Architecture undecided. Requirements rough. Existing system producing inconsistent results with no clear cause. That's the entry point.

Systems Thinking · Ambiguity · Technical Depth · Documentation · Handoffs

What it produces

A system with a defensible logic – documented, sourceable, buildable, and maintainable by someone other than the original designer.

Systems Thinking · Ambiguity · Technical Depth · Documentation · Handoffs

Features

Milestone

2019

Top-percentile GitHub contributor

[Source](#) ¹

Highlight

Imager

Rust / Node image optimization algorithm using brute force heuristics

[Source](#) ³

source :		39.00M (4 images)
kraken.io :		24M
jpegmini.com :		16M
compression.ai :		8.90M
imager :		4.20M

Achievement

728+ ★

GitHub stars on most popular authored OSS tool

[Source](#) ²

Review

Academic

"You are such a profound writer and thinker. It has been my privilege to be your instructor of record. One day I will say, I had him in my English class. I have such high hopes for you! Go conquer your world! You're awesome."

– Dr. Jim Birrell
[Research Project](#) ⁴

Early Career History

pre 2015

Space Monkey

Brief High School Internship

Worked under Tom Metge

2016

Uplynk / Verizon Digital Media Services

Worked as a Jr. Developer on the QA team.

Team lead: Asiel Brumfield.

Left to work on encrypted images.

2015

Galileo Processing

Galileo Processing

Worked under Robert Raver.

2020

2020 – UVU CS Grader

Utah Valley University

CS Grader – Computer Science

Bianca Ruiz

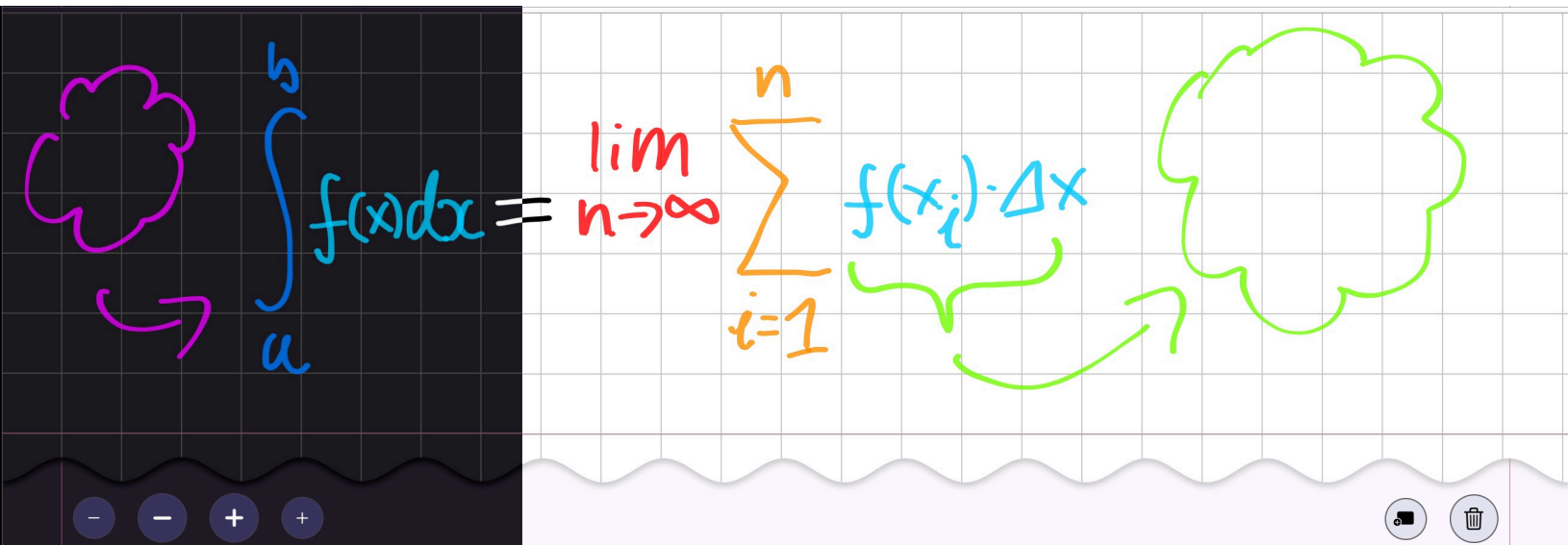
Overview links

1 <https://github.com/colbyn?tab=overview&from=2019-12-01&to=2019-12-31>

2 <https://github.com/imager-io/imager>

3 <https://github.com/imager-io/imager>

4 <https://github.com/colbyn/analog-computing-and-refs>



Keywords

Infrastructure

Data center operations · mainframe buildout · rack and stack · hardware provisioning · systems support · network documentation · structured cabling · Galileo processing · ticketing · incident response · physical layer · server configuration · storage · uptime · ops

Networks & Field

ISP founding · wireless infrastructure · WISP operations · last-mile · point-to-point · RF planning · subscriber management · routing · switching · TCP/IP · field installation · tower work · contracting · independent practice · sales · client acquisition · troubleshooting

Professional & Leadership

Cross-functional collaboration · stakeholder communication · project management · requirements gathering · deliverable ownership · process improvement · change management · knowledge transfer · vendor management · client-facing communication · scope definition · resource planning · risk assessment · quality assurance · agile · roadmap alignment · strategic planning · budget awareness · compliance · audit readiness

History

2014 · Summer Internship · Space Monkey · Decentralized P2P Cloud Computing · Infrastructure Provisioning · Systems Support
2015 · CompTIA Operations Specialist · Galileo Processing · Data Center Buildout · Hardware Provisioning · Network Documentation · Incident Response · Uptime & Ops
2015-2016 · Founder & Lead Systems Engineer · Independent WISP · Wireless Infrastructure · Last-Mile Deployment · RF Planning · Field Installation · Client Acquisition · Routing & Switching
2016 · QA Engineer, Media Systems · UpLynk / Verizon Digital Media Services · Video Streaming Pipelines · Encoding & Transcoding · HLS · CDN Delivery · Python Tooling · Throughput Optimization
2017 · Prototyping & Tooling Engineer · Eimg Startup · Video Processing Tooling · Rapid Prototyping · Multimedia Workflow Automation
2018 · Independent Software Engineer & Language Designer · Compiler Design · DSL Implementation · Rust · Haskell · Open Source Tooling
2019 · Independent Imaging Systems Developer · Imager Project · Image Processing Pipelines · FFmpeg Integration · VMAF Quality Metrics · Rust · Node.js · Performance Optimization
2020 · CS Student & Research Assistant · UVU · Academic Research · CS Grader · Subscript Publishing Toolchain · Swift · iPad Application Development
2021 · Independent DevOps & Cloud Consultant · AMI Automation · AWS Infrastructure · CI/CD Pipelines · Rust Systems Programming
2022 · Independent Researcher & Systems Developer · University of Utah · Philosophy, Writing & Rhetoric · Parser Development · Publishing Systems · Rust · Swift · Technical Documentation
2023 · Software Engineer, Publishing Systems · Subscript Publishing · iOS & macOS Development · Digital Ink & Vector Graphics · Custom Language Design · Cross-Platform Architecture
2024 · Independent Systems Engineer · Parser Combinators & AST Tooling · Dataset Generation Pipelines · Rust · Swift · Data Architecture
2025-present · Principal Consultant, Systems & Integration · Static Web Compiler · Markdown Rendering · LED & Low-Voltage Systems · DALI-2 · PoE · Technical Writing · Architecture & Client Consulting

Media & Streaming

Video streaming · multimedia · codec pipelines · encoding · transcoding · multi-bitrate · adaptive streaming · HLS · CDN · delivery infrastructure · UpLynk · Verizon Digital Media Services · latency · throughput · media processing · ingest · playback · format conversion

Low Voltage & Lighting

Class 2 wiring · NEC compliance · DC distribution · LED systems · driver architecture · DALI-2 · dimming · CCR · PWM · constant current · PoE · Zhaga · COB · photometry · circadian lighting · spectral design · low-voltage networking · system architecture · home run topology · power budgeting · field installation · commissioning

Soft Skills & Competencies

Communication · active listening · relationship building · conflict resolution · negotiation · presentation skills · executive presence · self-directed · initiative · ownership mentality · accountability · adaptability · resilience · growth mindset · async communication · deadline-driven · detail-oriented · results-oriented · critical thinking · decision-making · prioritization · time management · problem-solving